FEISHU

Feishu Quant Competition

March 22, 2026

Contents

1 Competition Overview **3**

2 Competition Timeline **3**
2.1 Registration . . . 3
2.2 Stage I: Competition . . . 3
2.3 Stage II: Competition Rankings and Award Ceremony . . . 4

3 Data Description **4**
3.1 Data Types . . . 4
3.2 Time Split . . . 5
3.3 Data Distribution . . . 5
3.4 Data Usage Policy . . . 5

4 Trading and Portfolio Rules **6**
4.1 General Constraints . . . 6
4.2 Execution Prices and Sell Mode . . . 6
4.3 Transaction Costs . . . 7
4.4 Lot Size and Rounding . . . 7
4.5 Missing Prices and Untradable Securities . . . 7

5 Submission **8**
5.1 File Format . . . 8
5.2 Column Definitions . . . 8
5.3 Filename Convention . . . 9
5.4 Duplicate Rows . . . 9
5.5 Example Submission . . . 9
5.6 Submission Portal . . . 10

6 Requirements for Shortlisted Teams **10**
6.1 Source Code . . . 10
6.2 Research Report . . . 10

7 Composite Score and Ranking Rules **11**
7.1 Composite Score . . . 11
7.2 Ranking Rules . . . 11

8 Awards and Prizes **12**
8.1 Prize Pool . . . 12
8.2 Career Development Opportunities . . . 12

9 Competition Rules **12**
9.1 Intellectual Property . . . 12
9.2 Data . . . 12
9.3 Academic Integrity and Fair Competition . . . 12

9.4 Final Interpretation 13

10 About Feishu China **13**

11 Academic and Research Partners **13**

12 Judging Panel **13**

1 Competition Overview

In contemporary quantitative investment practice, research delivers genuine long-term value only when it leads to implementable, executable, and reproducible portfolio strategies, rather than merely demonstrating statistical factor predictability. This quantitative trading competition is built around a realistic asset management setting and centers on the design, execution, and evaluation of long-only portfolio strategies in the A-share market.

In contrast to conventional factor competitions that focus on neutralized alpha factors or RankIC metrics, this competition neither requires style neutralization nor applies residualization based on risk models such as Barra. Instead, participants' investment capability is assessed directly through end-to-end portfolio backtesting results. Subject to strict trading rules and execution constraints, participants are required to construct long-only portfolios using high-frequency limit order book (LOB) data together with daily-frequency OHLCV market data, and their strategies are evaluated on a fully out-of-sample basis.

2 Competition Timeline

2.1 Registration

Participants must complete registration via the Feishu Virtual Campus website:

https://campus.feishu.ca/

In the registration website, participants must create a new account, then enter the enrollment key to enroll the competition:

`gYC5a4H47v5Mf9r`

Registration must be completed prior to the release of the out-of-sample data on May 28, 2026. If you have any questions, please contact:

`info@feishu.ca`

2.2 Stage I: Competition

Stage I of the competition will run from March 22, 2026 to June 1, 2026.

- Participants may compete either individually or in teams of up to three members.
- The in-sample data will be released at 12:00 PM (Beijing Time) on March 22, 2026.
- The out-of-sample data will be released at 12:00 PM (Beijing Time) on May 28, 2026.
- All submission materials must be submitted no later than 12:00 PM (Beijing Time) on June 1, 2026.
- During this stage, participating teams may use the data provided by the organizer to develop portfolio strategies and prepare their submissions in accordance with the competition rules.

2.3 Stage II: Competition Rankings and Award Ceremony

In Stage II, the competition rankings, as determined according to Section 7, will be announced in early June 2026.

The award ceremony will be announced separately and is scheduled to take place in the second half of 2026, with invited guests, judges, and contestants in attendance.

3 Data Description

3.1 Data Types

The organizer will provide participants with standardized market data from the Shanghai Stock Exchange, with all datasets strictly synchronized along the time dimension. Recently listed stocks are excluded from the sample. The total dataset size is approximately 3 GB. Specifically, the data consist of:

- **10-minute LOB data** * derived from second-level Shanghai Stock Exchange data and downsampled to a 10-minute frequency, including 10-level order book snapshots and quote information for market microstructure analysis and high-frequency feature construction. Table 1 summarizes the column structure.

Table 1: Column descriptions of 10-minute LOB data.

Column	Type	Description
`asset_id`	string	Unique asset identifier (e.g., `A000001`, `A000002`, …)
`trade_day_id`	string	Trading day identifier (e.g., `D001`, `D002`, …)
`time`	string	Intraday time slot (09:40:00–15:00:00, 24 unique values, 10-minute intervals)
`ask_price_`i	float32	Ask price at level i, level 1 = best ask
`bid_price_`i	float32	Bid price at level i, level 1 = best bid
`ask_volume_`i	float32	Ask volume at level i
`bid_volume_`i	float32	Bid volume at level i

- **Daily OHLCV data:** provided for the same stock universe and used for modeling lower-frequency trends, volatility, and market state characteristics. Table 2 summarizes the columns provided in the daily OHLCV dataset.

*The first snapshot is recorded at 09:40:00, capturing the completed 10-minute interval following the 09:30:00 market open.

Table 2: Column descriptions of daily OHLCV data.

Column	Type	Description
`asset_id`	string	Unique asset identifier (e.g., `A000001`, `A000002`, …)
`trade_day_id`	string	Trading day identifier (e.g., `D001`, `D002`, …)
`open`	float64	Opening price
`high`	float64	Daily high price
`low`	float64	Daily low price
`close`	float64	Closing price
`volume`	float64	Trading volume
`amount`	float64	Trading amount
`adj_factor`	float64	Price adjustment factor
`vwap_0930_0935`	float64	Volume-weighted average price over 09:30–09:35

3.2 Time Split

To ensure a rigorous and forward-looking evaluation framework, the dataset is explicitly partitioned into an in-sample period and an out-of-sample period. The in-sample period may be used for model training, parameter calibration, and strategy development, whereas the out-of-sample period is reserved solely for inference and must not be used for any form of training, validation, or hyperparameter tuning.

- **In-sample period:** D001 to D484.
- **Out-of-sample period:** D485 to D726.

3.3 Data Distribution

Upon successful registration, each participant will receive login credentials for the Feishu Virtual Campus platform. The datasets will be made available for download through the platform according to the schedule specified in Section 2.2.

3.4 Data Usage Policy

All datasets provided by the organizer are strictly for use in this competition only. Participants are prohibited from using the data for any commercial purpose, academic publication, or redistribution without prior written consent from the organizer. Participants must not share, transfer, or disclose any part of the datasets to individuals or teams outside their own registered team. Violation of this policy will result in immediate disqualification.

4 Trading and Portfolio Rules

4.1 General Constraints

The backtesting environment is designed to closely replicate real-world A-share market conditions. All trades must strictly follow the T+1 trading rule in the A-share market, meaning that stocks purchased on a given trading day may not be sold on the same day.

Each team begins the OOS period with an initial capital of RMB 50 million. At the end of each trading day, the portfolio must hold at least 10 stocks. Submissions that result in a portfolio holding fewer than 10 stocks on any trading day will be rejected and will not be evaluated.

4.2 Execution Prices and Sell Mode

In this section, `buy_percentage` and `sell_percentage` refer to the per-asset instructions submitted by participants, as formally defined in Section 5.2.

Buy (day t). All buy orders are executed on day t using the 5-minute opening VWAP:

- Buy price on day t: `vwap_0930_0935` (09:30–09:35 VWAP) available in the daily OHLCV dataset, see Table 2).

Sell (day t). Each team must choose exactly one sell mode for the entire OOS period, and the chosen mode must remain consistent across all submissions:

- **Sell-at-Open**: sell price on day t is `open` on day t.
- **Sell-at-Close**: sell price on day t is `close` on day t.

`sell_percentage` specified on day t is executed on the same day t at the selected sell point. If both buy and sell are specified on the same day, sell orders are processed before buy orders.

Buy execution order. Buy orders are executed sequentially in the order they appear in the submission file. For each buy order, the execution quantity is computed based on the remaining available cash at that point in time, after deducting all prior transaction costs. Participants must ensure that their submitted `buy_percentage` values are feasible given available cash, transaction costs, and lot size constraints. No normalization is applied to `buy_percentage`.

No leverage. If available cash is insufficient to execute a buy order (including all associated costs), the order receives no fill (0 shares). No partial fill or carry-over is applied.

4.3 Transaction Costs

All performance metrics are evaluated with transaction costs, which are deducted from cash at execution time. Let turnover = executed shares × execution price. The cost structure is as follows:

- **Commission** (both buy and sell): 0.01% (1 bp) of turnover, subject to a minimum of RMB 5 per order.
- **Stamp duty** (sell-side only): 0.05% (5 bps) of turnover.

Concretely:

$$\text{Buy cost} = \max(\text{turnover} \times 0.0001,\ 5),$$

$$\text{Sell cost} = \max(\text{turnover} \times 0.0001,\ 5) + \text{turnover} \times 0.0005.$$

4.4 Lot Size and Rounding

The minimum trade unit is 100 shares (1 lot), consistent with Shanghai Stock Exchange (SSE) trading rules.

Buy orders. The execution quantity is determined as follows:

1. Compute the theoretical share count by truncating `cash/price` toward zero.
2. Truncate the result toward zero to the nearest lot of 100 shares.
3. If the total cost (execution amount plus transaction costs) exceeds available cash, reduce the quantity by one lot at a time until the total cost fits within available cash.
4. If the final quantity is fewer than 100 shares, the order is treated as no fill.

Sell orders.

- **Full liquidation** (`sell_percentage` = 1): all sellable shares are sold. If this would leave a remaining position of fewer than 100 shares, that remainder is also sold in full.
- **Partial sell** (`sell_percentage` < 1): the sell quantity is computed from the current sellable shares and truncated toward zero to the nearest lot of 100 shares. Any odd-lot remainder is retained in the portfolio.

4.5 Missing Prices and Untradable Securities

If a required execution price (`vwap_0930_0935`, `open`, or `close`) is missing, or if an asset is untradable at the relevant execution point (e.g., due to suspension or no trades), the corresponding order is treated as no fill (0 shares) with no carry-over to future trading days.

5 Submission

5.1 File Format

During the out-of-sample (OOS) phase, each team must submit one CSV file containing exactly the following four column:

`trade_day_id, asset_id, buy_percentage, sell_percentage`

Rows where both `buy_percentage` = 0 and `sell_percentage` = 0 must be removed prior to submission.

Validation and rejection. Submissions that fail to meet any of the formatting requirements specified in this section—including incorrect column names, missing columns, invalid column types, or filenames that do not conform to the naming convention—will be automatically rejected without notice. No corrections or resubmissions will be accepted after the deadline.

Row ordering. The CSV file must be sorted in ascending order by trade_day_id (earliest to latest).

5.2 Column Definitions

Table 3 summarizes the four required columns in the submission file.

Table 3: Column definitions of the submission file.

Column	Type	Description
`trade_day_id`	string	Anonymized trading day identifier (e.g., `D001`, `D002`, ...). T+1 is defined by sequential ordering.
`asset_id`	string	Anonymized asset identifier in the format `A` + 6 digits (e.g., `A000001`, `A600000`). Must exactly match IDs in the official datasets; leading zeros must be preserved.
`buy_percentage`	float $\in [0, 1]$	Fraction of **available cash** allocated to buy this asset on day t at the buy execution price. Orders are executed sequentially in CSV row order; no normalization is applied. Note: this is a cash allocation, not a target portfolio weight.
`sell_percentage`	float $\in [0, 1]$	Fraction of **this asset's sellable shares** to sell on day t. Due to the **A-share T+1 rule**, shares bought on day t are not sellable until day $t+1$ (i.e., `D001` $\rightarrow$ `D002`), and therefore only shares carried from prior days are eligible for sale.

Decimal precision. `buy_percentage` and `sell_percentage` must be decimal numbers in $[0, 1]$ with at most 6 decimal places (e.g., `0.4`, `0.333333`). Values exceeding 6 decimal places will be truncated before normalization and execution are applied.

5.3 Filename Convention

The submission filename must include the team identifier and the chosen sell mode:

- `TEAMID_sell_open.csv`
- `TEAMID_sell_close.csv`

For example, a team with ID `T023` would submit either `T023_sell_open.csv` or `T023_sell_close.csv`.

5.4 Duplicate Rows

Each (`trade_day_id`, `asset_id`) pair must appear at most once per submission. If duplicate pairs are present, the organizer will retain only the last occurrence based on CSV row order and discard all earlier duplicates.

5.5 Example Submission

The following example illustrates a valid submission under Sell-at-Open mode. In Sell-at-Close mode, the only difference is that sell orders are executed at `close` instead of `open`.

Table 4: Example submission file (`T023_sell_open.csv`).

trade_day_id	asset_id	buy_percentage	sell_percentage
D001	A600000	0.40	0.00
D001	A000001	0.536473	0.00
D002	A600000	0.00	1.00
D002	A000001	0.00	0.487913

This submission is interpreted as follows:

- On `D001`: buy `A600000` and `A000001` at `vwap_0930_0935`, allocating 40% and 53.6473% of available cash, respectively.
- On `D002`: execute sell orders at `open`—sell 100% of sellable shares of `A600000` and 48.7913% of sellable shares of `A000001`. Shares bought on `D002` are not sellable on the same day due to the A-share T+1 rule.

5.6 Submission Portal

All submissions must be uploaded via the official submission portal provided by the organizer. Details of the submission method will be communicated to registered teams together with the out-of-sample data release on May 28, 2026. Each team is permitted to submit exactly once; resubmissions will not be accepted. Teams are strongly advised to verify their submission file before uploading.

6 Requirements for Shortlisted Teams

At a later stage, the organizer will notify the top 10 teams of their shortlisted status. Shortlisted teams are required to submit the following additional materials within the specified deadline:

6.1 Source Code

The submitted code must be capable of reproducing the corresponding out-of-sample results under the data and runtime environment specified by the organizer, and must strictly comply with the usage rules for the out-of-sample data. In particular, the out-of-sample data must not be used for any form of model training, parameter tuning, or strategic structure adjustment.

6.2 Research Report

Each shortlisted team must submit a 10–15 page research report in PDF format describing its portfolio strategy. The report must be complete in structure, rigorous in methodology, and clear in presentation. It is recommended that the report include, at a minimum, the following sections:

- **Abstract**: a concise summary of the strategy, including the research objective, data used, core methodology, main empirical findings, and key conclusions.
- **Introduction**: research background and literature review.
- **Factor Construction**: detailed description of data preprocessing, factor design, relevant formulas, and methodological rationale.
- **Empirical Analysis**: comprehensive presentation of backtesting results and performance analyses.
- **Innovation and Limitations**: a discussion of the novelty and originality of the proposed strategy, together with a balanced assessment of its potential limitations, assumptions, implementation constraints, and sources of model risk.
- **Conclusion**: a clear summary of the main findings, overall effectiveness of the strategy, and the key insights drawn from the competition, together with possible directions for future improvement or extension.

7 Composite Score and Ranking Rules

7.1 Composite Score

The organizer will convert each evaluation metric into a percentile ranking across all submitted strategies, defined as follows:

- CAGR_{pct}: Percentile ranking of the Compound Annual Growth Rate (CAGR).

  Definition of the CAGR: The annualized growth rate of the portfolio over the evaluation period, defined as

  $$\text{CAGR} = \left(\frac{V_T}{V_0}\right)^{\frac{1}{T}} - 1,$$

  where V_0 and V_T denote the initial and final portfolio values, respectively, and T is the length of the evaluation period measured in years.

- SR_{pct}: Percentile ranking of the annualized Sharpe Ratio (SR).

  Definition of the Annualized SR: The annualized risk-adjusted return of the portfolio, defined as

  $$\text{SR} = \frac{\mu(R_t - R_f)}{\sigma(R_t - R_f)}\sqrt{N},$$

  where R_t denotes the portfolio return at time t, R_f denotes the risk-free rate, $\mu(\cdot)$ denotes the mean of excess returns, $\sigma(\cdot)$ denotes the standard deviation of excess returns, and N is the number of trading periods in one year.

- MDD_{pct}: Percentile ranking of $-\text{MDD}$, so that a lower drawdown yields a higher score.

  Definition of the MDD: The Maximum Drawdown (MDD) is the maximum observed percentage decline in portfolio value from a historical peak to a subsequent trough over the evaluation period, defined as

  $$\text{MDD} = \max_{0 \le t \le T}\left(\frac{P_t - V_t}{P_t}\right),$$

  where V_t denotes the portfolio value at time t, and $P_t = \max_{0 \le s \le t} V_s$ is the running peak portfolio value up to time t.

The final composite score is defined as:

$$\text{Score} = 0.45 \cdot \text{CAGR}_{pct} + 0.30 \cdot \text{SR}_{pct} + 0.25 \cdot \text{MDD}_{pct}.$$

7.2 Ranking Rules

- All submitted strategies will be ranked in descending order of the composite score.
- The composite score is the sole basis for the final ranking.
- In the event of a tie in composite score, strategies will be ranked by CAGR first and then by SR.

8 Awards and Prizes

8.1 Prize Pool

A total prize pool of **RMB 40,000** will be awarded in this competition, with the distribution as follows:

- **First Prize**: RMB 10,000 (1 team).
- **Second Prize**: RMB 7,500 (2 teams).
- **Third Prize**: RMB 5,000 (2 teams).
- **Innovation Prize**: RMB 2,500 (2 teams).

8.2 Career Development Opportunities

In addition to monetary awards, outstanding winners will be granted interview opportunities with top institutions in the fields of financial technology, asset management, AI infrastructure, data science, and large language models (LLM). The potential annual compensation for such opportunities may reach up to **RMB 1 million**.

9 Competition Rules

9.1 Intellectual Property

The intellectual property rights to all submissions, including code, reports, and other materials, shall remain with the participating teams. The organizer reserves the non-commercial right to evaluate, display, and promote all submitted works. For outstanding prize-winning strategies, the organizer shall have the right of first negotiation for potential commercial collaboration under equivalent conditions.

9.2 Data

The data provided in this competition are intended exclusively for academic research and competition purposes. Participants are not permitted to use the data for any commercial purpose, including but not limited to product development, commercial analysis, resale, redistribution, or any other profit-oriented activities, unless explicit written authorization has been obtained from Feishu.

9.3 Academic Integrity and Fair Competition

Any form of plagiarism, cheating, or unauthorized code sharing is strictly prohibited. Any team found to have engaged in such conduct will be immediately disqualified from the competition.

9.4 Final Interpretation

The organizer reserves the final right of interpretation of all rules and regulations of this competition.

10 About Feishu China

In September 2024, Feishu China was officially established in Putuo District, Shanghai. Leveraging its extensive experience in global academic, industrial, and governmental collaboration, Feishu China is committed to establishing a world-class platform for mathematical research and innovation, advancing mathematical sciences while fostering new quality productive forces.

Feishu China maintains deep collaboration with internationally renowned institutions including the University of Toronto, ETH Zurich, and Columbia University, with a strategic focus on the intersection of mathematics and financial risk. The Feishu Quant Competition represents a key initiative by Feishu China to advance the integration of mathematics and quantitative finance, and to cultivate high-caliber quantitative research talent.

11 Academic and Research Partners

The competition is supported by academic and research partners with strong expertise in mathematical finance, quantitative research, risk management, and financial engineering.

- Mathematical Finance Program, University of Toronto.
- RiskLab, University of Toronto.
- Mathematical Finance Department, Technical University of Munich.
- Joint MSc Program in Quantitative Finance, ETH Zurich and the University of Zurich.

12 Judging Panel

The competition will be evaluated by a panel of experts from academia and industry, with expertise in quantitative finance, asset management, financial technology, artificial intelligence, and data-driven investment research. The judging panel will be responsible for reviewing submissions, assessing research quality and practical relevance, and determining the final awards in accordance with the competition rules.